

2022 Year 11 Yearly Exam - Mathematics Standard - Worked Solutions

1. **B**

From the reference sheet, the volume of a sphere is

$$V = \frac{4}{3}\pi r^3, \text{ so } V = \frac{4}{3} \times \pi \times 5^3$$
$$V = 523.59877 \dots \text{ cm}^3$$
$$V = 523.60 \text{ cm}^3 \text{ (2 dec. places)}$$

Hence, the correct option is B.

2. **C**

On Sunday Darren earned $\$55 \times 9 = \495 .

So, on Saturday Darren earned $\$855 - \$495 = \$360$.

Hours worked on Saturday = $\$360 \div \$45 = 8$.

Hence, the correct option is C.

3. **D**

In option A, the area of the parallelogram is

$$A = \text{base} \times \text{perpendicular height} = 6 \times 7 = 42 \text{ cm}^2.$$

In option B, the area of the rhombus is

$$A = \frac{1}{2}xy = \frac{1}{2} \times 6 \times 14 = 42 \text{ cm}^2.$$

In option C, the area of the trapezium is

$$A = \frac{h}{2}(a + b) = \frac{6}{2}(5 + 9) = 42 \text{ cm}^2.$$

In option D, the area of the Kite is

$$A = \frac{1}{2}xy = \frac{1}{2} \times 6 \times 12 = 36 \text{ cm}^2.$$

So, the smallest area is the area of the kite.

Hence, the correct option is D.

4. **A**

Each interval on the vertical axis this cumulative frequency histogram represents 3 members.

Now, 9 members trained for 4 hours and 21 members trained for 5 hours or less this means the number of members who trained for 5 hours is the difference which is $21 - 9 = 12$.

Hence, the correct option is A.

5. **D**

The difference in time zones is based on the difference in the longitude of the locations. City A has longitude 82°E and City B has longitude 23°W , so the difference is $82^\circ + 23^\circ = 105^\circ$.

$105^\circ \div 15^\circ = 7 \text{ hours}$. As City A is East of City B this means it is 7 hours ahead of City B.

So, when it is 10 pm Monday in city B it will be 5 am Tuesday in City A.

Hence, the correct option is D.

6. **B**

In the first 6 days she spent $\$125 \times 4 = \500 .

Over the 10 days she spent $\$140 \times 10 = \1400 .

This means over the last 6 days she spent

$$\$1400 - \$500 = \$900.$$

So, her average spending over the last 6 days is $\$900 \div 6 = \150 .

Hence, the correct option is B.

7. **A**

The probability that the first marker selected is blue is $\frac{4}{9}$.

If the first marker selected is blue, then there are 3 blue markers amongst the remaining 8 markers, so the probability of the second marker being blue is $\frac{3}{8}$.

So, the probability that Jacob selects 2 blue markers is $\frac{4}{9} \times \frac{3}{8}$.

Hence, the correct option is A

8. **C**

The area of the two triangles is

$$A_1 = 2 \times \frac{1}{2} \times 8 \times 6 = 48 \text{ cm}^2.$$

The area of the front rectangle is

$$A_2 = 10 \times 12 = 120 \text{ cm}^2.$$

The area of the right side rectangle is

$$A_3 = 8 \times 12 = 96 \text{ cm}^2.$$

The area of the left side rectangle is

$$A_4 = 6 \times 12 = 72 \text{ cm}^2$$

So, the total surface area is

$$A = 48 + 120 + 96 + 72 = 336 \text{ cm}^2.$$

Hence, the correct option is C

9. **D**

$$\text{Absolute error} = \frac{1}{2} \text{ of the precision} = \pm 0.5 \text{ cm}$$

$$\text{Percentage error} = \frac{\text{absolute error}}{\text{measurement}} \times 100$$

$$= \frac{\pm 0.5}{400} \times 100 = \pm 0.125 \%$$

Hence, the correct option is D.

10. **B**

As Lachlan gave his brother 200 mangoes, he has 3800 mangoes left. Each week he sells 250 mangoes this means in p weeks he will sell $250p$.

So, the number of mangoes he still has after p weeks is

$$N = 3800 - 250p.$$

Hence the correct option is B.

11. C

The line cuts the y axis at -2 .

As the line is sloping down the gradient is negative.

Also, as the vertical rise is 2 units and a horizontal run of 5 units then the gradient of this line is $-\frac{2}{5}$.

As the equation of a line is $y = mx + b$ then its equation is $y = -\frac{2}{5}x - 2$

Hence, the correct option is C.

12. C

In one second, he was given $\frac{9}{4}$ drops.

So, in one minute he was given

$$\frac{9}{4} \times 60 = 135 \text{ drops.}$$

In one hour, he was given

$$135 \times 60 = 8\,100 \text{ drops.}$$

In 8 hours, he was given

$$8\,100 \times 8 = 64\,800 \text{ drops.}$$

Now, this person was given 0.8 L or 800 mL in the 8 hours

this means the rate of fluid is

$$64\,800 \text{ drops} \div 800 \text{ mL} = 81 \text{ drops/mL.}$$

Hence, the correct option is C

13. A

Let P be the amount that Kylie won which is the same as the amount invested.

$$P + \text{simple interest} = \$64\,800.$$

Also, simple Interest = $P \times 0.04 \times 5$ this means

$$P + P \times 0.04 \times 5 = 64\,800$$

$$P + P \times 0.2 = 64\,800$$

$$1.2 \times P = 64\,800$$

$$P = 64\,800 \div 1.2 \text{ So, } P = 54\,000.$$

Hence, the correct option is A

14. D

In bag A, there are x lollies in total so the number of honey lollies in this bag is $\frac{4}{9}x$.

Also, in bag B, there are x lollies in total so the number of honey in this bag is $\frac{2}{3}x$.

As there are 6 more honey lollies in bag B than in bag A this means $\frac{2}{3}x = \frac{4}{9}x + 6$.

To solve this equation, we multiply both sides by 9, we get $6x = 4x + 54$ that is $2x = 54$ So, $x = 27$.

Hence, the correct option is D.

15. A

Let J be John's age, by using the formula, we get

$$2 = \frac{J \times 30}{150} \text{ this indicates } 30J = 300$$

that is $J = 10$. So, John is 10 months old.

Let T be Tania's age, by using the formula, we get

$$3 = \frac{T \times 30}{150} \text{ this indicates } 30T = 450$$

that is $T = 15$. So, Tania is 15 months old.

Therefore, Tania is 5 months older than John.

Hence, the correct option is A.

Question 16

The value of the boat depreciates by \$ 950 each year for 3 years, so the total depreciation is

$$\$950 \times 3 = \$2\,850.$$

So, the purchase price was

$$\$7\,200 + \$2\,850 = \$10\,050.$$

Question 17

$$\begin{aligned} \text{a) Mean} &= \frac{350+270+332+460+342+280+282}{7} = \frac{2316}{7} \\ &= 330.86 \text{ correct to 2 decimal places.} \end{aligned}$$

This could also have been found using the statistics mode on the calculator, and so no working would be necessary.

b) Using the statistics mode on the calculator, the standard deviation is 60.77 correct to 2 decimal places.

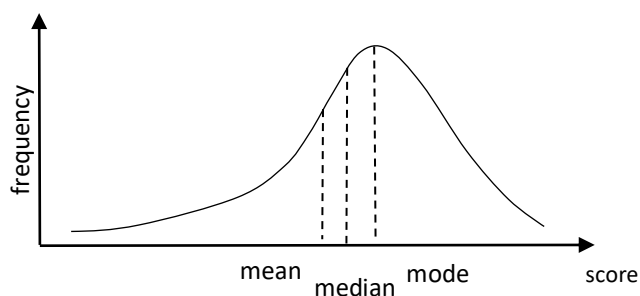
Question 18

For a 4 hour job, the cost would be

$$\$50 + 4 \times \$80 = \$370.$$

The 10% discount will be \$37, so the electrician would charge $\$370 - \$37 = \$333$.

Question 19



A negatively skewed distribution has a long tail at the lower scores end (left end) and the mean to the left of the median and the mode to the right of the median.

Question 20

Sara's car uses 9.2 litres for 100 km and by multiplying by 1.2 we can see that it uses 11.04 litres for 120 km.

So, Sara's cost is $11.04 \times \$1.75 = \19.32 .

Rachel's car uses 7.6 litres for 100 km and by multiplying by 1.2 we can see that it uses 9.12 litres for 120 km.

Rachel's cost is $9.12 \times \$1.75 = \15.96

Hence, Sara's paid $\$19.32 - \$15.96 = \$3.36$ more than Rachel for fuel.

Alternative method

Sara's uses $11.04 - 9.12 = 1.92$ L more petrol.

Hence, Sara paid $1.92 \times \$1.75 = \3.36 more than Rachel for petrol.

Question 21

City Q is 7 hours ahead of City P.

When the plane left City Q the time was 1:20 am Monday, so the time in City P would 7 hours behind that is 6:20 pm on Sunday.

The flight took 10 hours, so it would arrive in City P at 4:20 am on Monday.

Question 22

The deposit is 10% of $\$12\,000 = \$1\,200$.

The Interest is calculated on the balance still owed which is $\$12\,000 - \$1\,200 = \$10\,800$.

As he has 5 months interest free then he will pay interest only 15 months not 20 months.

Simple interest charged will be

$$I = \$10\,800 \times 0.05 \times \frac{15}{12} = \$675.$$

The total money he needs to pay will be

$$\$10\,800 + \$675 = \$11\,475.$$

Hence, his monthly repayment is

$$\$11\,475 \div 20 = \$573.75.$$

Question 23

There are 6 fortnights in 12 weeks, so Darren's total income will be $6 \times \$2\,550 = \$15\,300$

His expenses over 12 weeks will be $6 \times \$1\,580 = \$9\,480$.

Hence, Darren will be able to save

$$\$15\,300 - \$9\,480 = \$5\,820.$$

Question 24

Jane's weekly salary is $\$90\,000 \div 52 = \$1\,730.77$

Her four weeks normal salary is

$$\$1\,730.77 \times 4 = \$6\,923.08$$

Her holiday loading will be

$$\$6\,923.08 \times 0.175 = \$1\,211.54$$

Her total holiday pay will be

$$\$6\,923.08 + \$1\,211.54 = \$8\,134.62.$$

Question 25

a) The interquartile range is $18 - 12 = 6$.

As the mean is 2 more than twice the interquartile range this indicates that the mean is $2 \times 6 + 2 = 14$.

b) From the reference sheet the lowest score before having an outlier is

$$Q_1 - 1.5 \times IQR = 12 - 1.5 \times 6 = 3 \text{ and}$$

the highest score before having an outlier is

$$Q_3 + 1.5 \times IQR = 18 + 1.5 \times 6 = 27.$$

The maximum range could be $27 - 3 = 24$.

Question 26

As the profit P of a chicken shop varies directly with the number N of chickens sold then $P = kN$, where k is a constant.

By substituting $N = 28$ and $P = 238$, we get

$$238 = k \times 28 \text{ that is } \frac{238}{28} = k \text{ So, } k = 8.5.$$

Therefore $P = 8.5N$.

Hence, when $N = 85$, $P = 8.5 \times 85 = \$722.50$.

Question 27

The patient received 170 mL in 1 hour and 25 minutes or in 85 minutes this means he received 1 mL in

$$\frac{85}{170} \text{ minutes.}$$

Hence, this patient will receive 238 mL in

$$238 \times \frac{85}{170} = 119 \text{ minutes or in 1 hour and 59 minutes.}$$

Question 28

a) From the reference sheet, the area of the sector is

$$A = \frac{\theta}{360^\circ} \times \pi r^2$$

$$A = \frac{60^\circ}{360^\circ} \times \pi \times 20^2 \approx 209.44 \text{ cm}^2 \text{ (2 dec. pl.)}$$

b) Using Pythagoras' Theorem in triangle ABC, we get

$$BC = \sqrt{20^2 - 12^2} = 16 \text{ cm.}$$

Area of the triangle is

$$A_{\text{Triangle}} = \frac{1}{2} \times 12 \times 16 = 96 \text{ cm}^2$$

The cross section area of the prism is

$$A_{\text{cross section}} = 209.44 + 96 = 305.44 \text{ cm}^2$$

Hence, the volume of the prism is

$$V = 305.44 \times 10 = 3054.4 \text{ cm}^3.$$

Question 29

a) As Daniel's allowable deductions are 5% of his gross income then his taxable income must be 95% of his gross income this means his gross income is $\$98\,325 \div 95 \times 100 = \$103\,500$.

b) Daniel's tax on his taxable income is

$$\text{Tax} = (\$98\,325 - \$45\,000) \times 0.325 + 5092 = \$22\,422.63$$

c) Daniel's Medicare levy on his taxable income is

$$0.02 \times \$98\,325 = \$1\,966.50$$

d) Daniel must pay the tax office a total of

$$\$22\,422.63 + \$1\,966.50 = \$24\,389.13$$

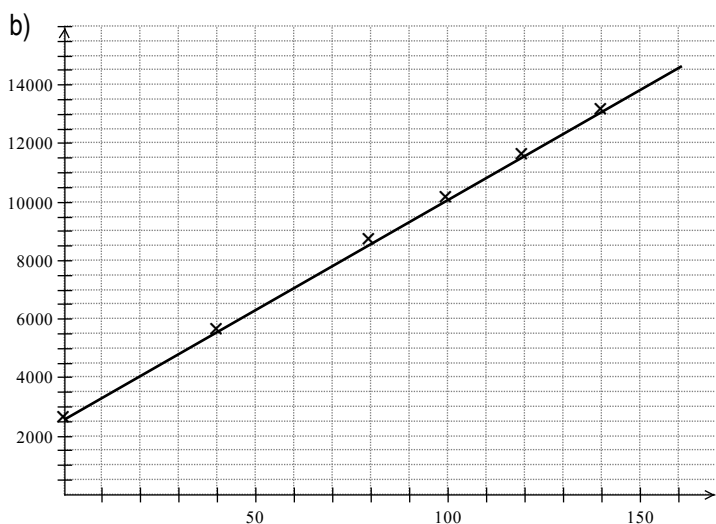
As he still needs to pay the tax office \$725 then he must have paid through PAYG tax throughout the last financial year

$$\$24\,389.13 - \$725 = \$23\,664.13$$

Question 30

a)

Number of Guests	0	40	80	100	120	140
Total cost in dollars	2500	5500	8500	10000	11500	13000



c) The cost is found by multiplying the number of guests N by \$75, and then adding \$2 500.

$$\text{So, } C = 75 \times N + 2500 \text{ or } C = 75N + 2500$$

d) When $N = 200$, $C = \$75 \times 200 + \2500

$$C = \$17\,500.$$

Hence, the cost is \$17 500 for 200 guests.

Question 31

As 1 kilocalorie = 4.184 Kilojoules then

$$61.9 \text{ kilocalories} = 61.9 \times 4.184 = 258.9896 \text{ Kilojoules.}$$

As 1 gram of fat contains 37 Kilojoules then

$$258.9896 \text{ Kilojoules} = 258.9896 \div 37 = 6.9997 \dots \text{ grams of fat.}$$

Hence, the can of corned beef contains approximately 7 grams of fat.

Question 32

The time from 12:10 am until 2:40 am there is 2.5 hours. This means the time needed for Patrick's BAC to reach zero after his last drink is 2.5 hours.

By substituting 2.5 in the formula :

$$\text{Number of hours for BAC to reach zero} = \frac{\text{BAC}}{0.015},$$

$$\text{we get } 2.5 = \frac{\text{BAC}}{0.015} \text{ so } \text{BAC} = 2.5 \times 0.015 = 0.0375.$$

Now, the of standard drinks that Patrick consumed is

$$N = 2 \times 1.9 = 3.8.$$

By substituting $\text{BAC}_{\text{Male}} = 0.0375$, $N = 3.8$ and Patrick's mass $M = 82 \text{ kg}$ in

$$\text{BAC}_{\text{Male}} = \frac{10N - 7.5H}{6.8M}, \text{ we get}$$

$$0.0375 = \frac{10 \times 3.8 - 7.5H}{6.8 \times 82} \text{ that is } 0.0375 = \frac{38 - 7.5H}{557.6}$$

By Multiplying by 557.6 on both sides, we get

$$20.91 = 38 - 7.5H \text{ this means } -17.09 = -7.5H$$

$$\text{So, } H = 2.27866 \dots h \approx 2 \text{ hours } 17 \text{ minutes.}$$

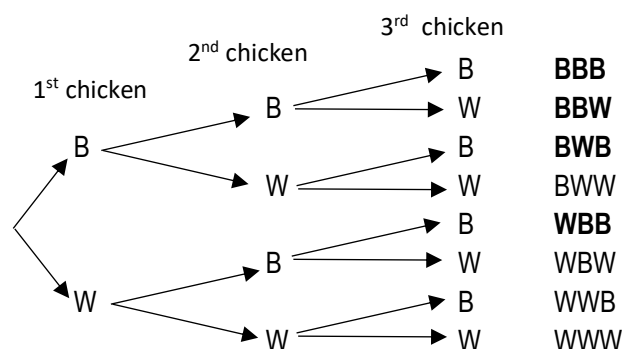
As Patrick stopped drinking at 12:10 am then he started to drink 2 hours 17 minutes before this time.

Now, 2 hours before 12:10 am is 10:10 pm and

17 minutes before 10:10 pm is 9:53 pm.

Hence, Patrick started his first drink at 9:53 pm.

Question 33



To select more brown chickens than white chickens then we must select 2 or 3 brown chickens.

The probability of 3 brown chickens is

$$P(\text{BBB}) = \frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} = \frac{1}{30}.$$

The probability of 2 brown chickens is

$$P(\text{BBW}) = \frac{4}{10} \times \frac{3}{9} \times \frac{6}{8} = \frac{1}{10}.$$

$$P(\text{BWB}) = \frac{4}{10} \times \frac{6}{9} \times \frac{3}{8} = \frac{1}{10}.$$

$$P(\text{WBB}) = \frac{6}{10} \times \frac{4}{9} \times \frac{3}{8} = \frac{1}{10}.$$

Hence, the probability of at least 2 brown chickens is

$$P = \frac{1}{30} + \frac{1}{10} + \frac{1}{10} + \frac{1}{10} = \frac{10}{30} = \frac{1}{3}.$$

Question 34

As the ratio of mango juice to water is 6 : 1 this means $\frac{6}{7}$ of the 210 mL of drink is mango juice. So, the amount of mango juice originally in the cup is

$$\frac{6}{7} \times 210 = 180 \text{ mL}.$$

After Karen added only water to the cup this indicates that the amount of mango juice remains the same and

180 mL now represents $\frac{9}{13}$ of the new mixture.

So, the total amount of the mixture is

$$180 \div 9 \times 13 = 260 \text{ mL}.$$

Therefore, the amount of water added is

$$260 - 210 = 50 \text{ mL}.$$

Alternative method

As the ratio of mango juice to water is 6 : 1 this means $\frac{6}{7}$ of the 210 mL of drink is mango juice. So, the amount of mango juice originally in the cup is

$\frac{6}{7} \times 210 = 180 \text{ mL}$ and the amount of water is

$$\frac{6}{7} \times 210 = 180 \text{ mL and the amount of water is}$$

$$210 - 180 = 30 \text{ mL}.$$

The new ratio is 9 : 4 and the amount of mango juice is

unchanged at 180 mL this means 9 parts represents

180 mL that is 1 part represents 20 mL.

As water is now 4 parts then the water is

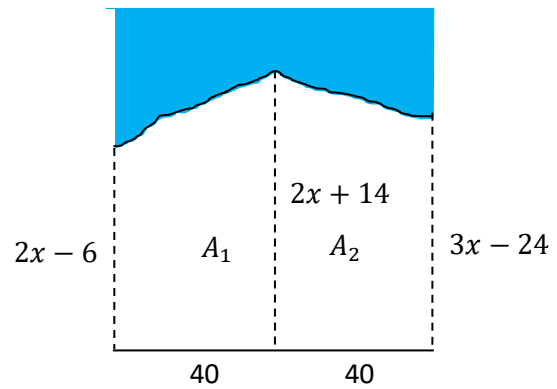
$$4 \times 20 = 80 \text{ mL}.$$

Hence, the amount of water added is

$$80 - 30 = 50 \text{ mL}.$$

Question 35

a)



Area of the first trapezium

$$\begin{aligned} A_1 &= \frac{40}{2} (2x - 6 + 2x + 14) \\ &= 20 (4x + 8) \\ &= 80x + 160 \end{aligned}$$

Area of the second trapezium

$$\begin{aligned} A_2 &= \frac{40}{2} (3x - 24 + 2x + 14) \\ &= 20 (5x - 10) \\ &= 100x - 200 \end{aligned}$$

So, the total area is

$$A = 80x + 160 + 100x - 200$$

$$A = 180x - 40$$

b) We are given that $\frac{1}{4} \times A = 50x - 150$ this means $A = 200x - 600$.

We now have 2 different expressions for A, so

$$200x - 600 = 180x - 40$$

$$20x = 560 \text{ that is } x = 28.$$

As $x = 28$ then $A = 200 \times 28 - 600$

$$A = 5000 \text{ m}^2$$

Hence, Melanie's land has an area of 5000 m^2 .

Alternative method

$$\frac{1}{4} \times A = \frac{1}{4} (180x - 40) = 45x - 10.$$

Now, as $\frac{1}{4} \times A = 50x - 150$ this means

$$45x - 10 = 50x - 150$$

$$5x = 140 \text{ that is } x = 28.$$

As $x = 28$ then $A = 200 \times 28 - 600$

$$A = 5000 \text{ m}^2$$

Hence, Melanie's land has an area of 5000 m^2 .